Deep Learning: The Perspective of Generalization

Bias–Variance Tradeoff, Double Descent

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March 26, 2025

Outline

1. Main Question and Introducing Topics
2. Motivation
3. Double Descent on Overparameterization
4. Double Descent on Training Dynamics
5. Conclusion
6. References

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1. Main Question and Introducing Topics
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Main Question

Main Question

Why doesn't deep learning overfit as badly as classical theory suggests?

- Goal of this talk: understand this gap through three steps.

1. Classical view: overfitting and bias–variance tradeoff
2. Overparameterization and double descent
3. Training dynamics and double descent

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Classical Regime

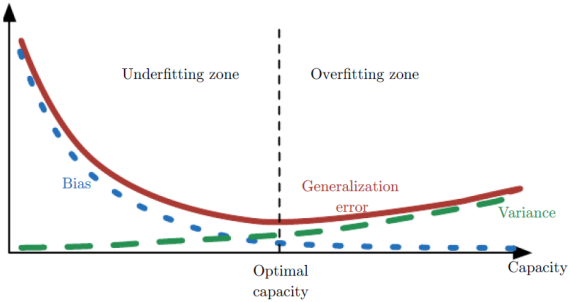

Figure 1: Schematic illustration of model capacity vs error.

- As model capacity increases, bias decreases but variance increases.
- Their sum produces the familiar U-shaped test error curve.

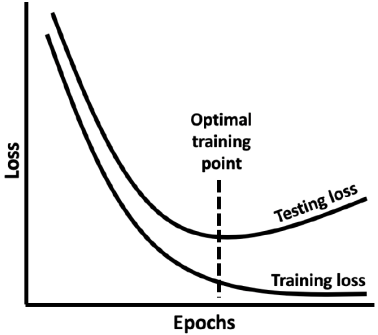

Figure 2: Schematic illustration of train vs test loss.

- Empirical view: training loss decreases, while test loss eventually increases.

Classical Regime

- **Decomposition:**

$$TestError = Bias^2 + Variance.$$

- **Underfitting:** typically associated with high bias and low variance, or equivalently, both training and test errors are high.
- **Overfitting:** typically associated with low bias but high variance, or equivalently, training error is low while test error is high.

Interpolation

A model $\hat{f}$ is said to *interpolate* the training data (x_i, y_i)'s if it fits every training sample exactly, i.e.,

$$\hat{f}(x_i) = y_i \qquad \text{for all } i = 1, \ldots, n.$$

Classical Implication

If a model interpolates noisy training data, that is usually interpreted as a warning sign:

$$\text{interpolation} \approx \text{high variance} \approx \text{poor generalization}.$$

But modern deep learning often violates this expectation.

Modern Regime: Double Descent

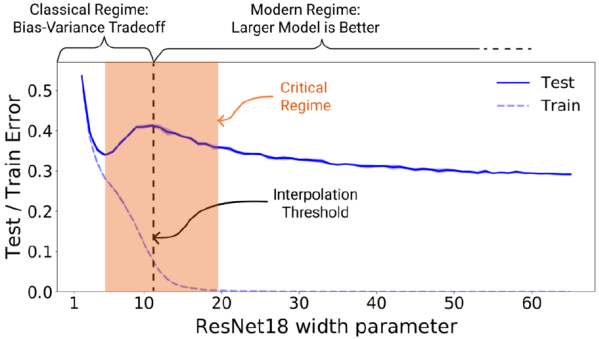

Figure 3: Example of double descent

- Near the interpolation threshold, test error can sharply increase.
- Beyond that threshold, larger models can generalize better again.
- Thus interpolation is not necessarily the end of generalization.

Modern Regime: Double Descent

- In modern deep learning, zero or near-zero training error often coexists with good generalization.
- Hence

$$\text{interpolation} \not\Rightarrow \text{overfitting}.$$

Model	Type	Reported size
GPT-3	LLM	175B parameters
PaLM	LLM	540B parameters
Llama 3.1	LLM	405B parameters
Stable Diffusion v1.x	Text-to-image diffusion	860M UNet (+123M text encoder)
Stable Diffusion XL / Turbo	Text-to-image diffusion	3.5B parameters

Table 1: Examples of large-scale generative models used in practice.

Modern Regime: Double Descent

Figure 4: Output of Stable Diffusion Model

Modern models often operate far beyond the classical regime.

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From Neural Networks to a Linear Model

We start from a nonlinear model

$$\mathbb{E}(y_i \mid z_i) = f(z_i; \theta), \qquad i = 1, \ldots, n,$$

where

$$y_i \in \mathbb{R}, \qquad z_i \in \mathbb{R}^d, \qquad \theta \in \mathbb{R}^p.$$

Assume training stays close to a random initialization $\theta_0 \in \mathbb{R}^p$, and define

$$\beta := \theta - \theta_0.$$

Then the first-order expansion around θ_0 with $f(z; \theta_0) \approx 0$ gives

$$\mathbb{E}(y_i \mid z_i) \approx \nabla_\theta f(z_i; \theta_0)^\top \beta, \qquad i = 1, \ldots, n.$$

Linearization at initialization

$$\mathbb{E}(y_i \mid z_i) = x_i^\top \beta, \qquad x_i := \nabla_\theta f(z_i; \theta_0) \in \mathbb{R}^p.$$

Linear Model and Min-Norm Interpolator

Assume we observe training data

$$(x_i, y_i), \qquad i = 1, \ldots, n,$$

with

$$x_1, \ldots, x_n \overset{\text{i.i.d.}}{\sim} P_x, \qquad \varepsilon_1, \ldots, \varepsilon_n \overset{\text{i.i.d.}}{\sim} P_\varepsilon, \qquad x_i \perp \varepsilon_i,$$

where P_x has mean $\mathbf{0}$ and covariance Σ, and P_ε has mean 0 and variance σ^2.

$$y_i = x_i^\top \beta + \varepsilon_i, \qquad i = 1, \ldots, n. \tag{1}$$

In matrix form,

$$y = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}. \tag{2}$$

When $p \gg n$, there are typically many interpolating solutions b satisfying

$$Xb = y.$$

Minimum-norm interpolator

Among all interpolating vectors, define

$$\hat{\boldsymbol{\beta}} := \arg\min_b \{\|b\|_2 : Xb = y\} = (X^\top X)^+ X^\top y,$$

where $(\cdot)^+$ denotes the Moore–Penrose inverse.

Gradient Descent Selects the Min-Norm Interpolator

Proposition 1 (Hastie et al. (2019), Proposition 1)

Initialize $\boldsymbol{\beta}^{(0)} = \mathbf{0}$, *and run gradient descent on the least-squares loss*

$$L(\boldsymbol{\beta}) = \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2,$$

with iterates

$$\boldsymbol{\beta}^{(k)} = \boldsymbol{\beta}^{(k-1)} + tX^\top\big(\mathbf{y} - X\boldsymbol{\beta}^{(k-1)}\big), \qquad k = 1, 2, 3, \ldots, \tag{3}$$

where

$$0 < t \leq \frac{1}{\lambda_{\max}(X^\top X)}. \tag{4}$$

Then

$$\lim_{k\to\infty} \boldsymbol{\beta}^{(k)} = \hat{\boldsymbol{\beta}}.$$

Why Gradient Descent Stays in the Row Space

Consider gradient descent updates (eq.(3)).

- Each increment(second term of RHS) has the form

$$X^\top v$$

  for some $v \in \mathbb{R}^n$.
- Hence every update direction lies in

$$\mathrm{col}(X^\top) = \mathrm{row}(X).$$

- Since

$$\beta^{(0)} = \mathbf{0} \in \mathrm{row}(X),$$

  all iterates remain in $\mathrm{row}(X)$.

The error evolves as

$$\beta^{(k+1)} - \beta^\star = (I - tX^\top X)(\beta^{(k)} - \beta^\star).$$

where $\beta^\star$ is a solution of least square loss.
Thus the step size must keep the eigenvalues of $I - tX^\top X$ within the unit disk. A sufficient condition is (4) which ensures stable convergence.

Why the Row-Space Solution is the Min-Norm Interpolator

Suppose $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ both interpolate the data:

$$X\boldsymbol{\beta}_1 = \mathbf{y}, \qquad X\boldsymbol{\beta}_2 = \mathbf{y}.$$

Then

$$X(\boldsymbol{\beta}_1 - \boldsymbol{\beta}_2) = \mathbf{0},$$

so their difference lies in $\mathrm{null}(X)$.

Thus any interpolating solution can be written as

$$\boldsymbol{\beta} = \boldsymbol{\beta}_{\mathrm{row}} + \boldsymbol{\beta}_{\mathrm{null}},$$

where

$$\boldsymbol{\beta}_{\mathrm{row}} \in \mathrm{row}(X), \qquad \boldsymbol{\beta}_{\mathrm{null}} \in \mathrm{null}(X).$$

We have the orthogonal decomposition

$$\|\boldsymbol{\beta}\|_2^2 = \|\boldsymbol{\beta}_{\mathrm{row}}\|_2^2 + \|\boldsymbol{\beta}_{\mathrm{null}}\|_2^2.$$

The minimum norm is achieved when

$$\boldsymbol{\beta}_{\mathrm{null}} = \mathbf{0}.$$

Therefore the unique interpolating solution in $\mathrm{row}(X)$ is exactly the min-norm interpolator.

Prediction Risk

For a test point $x_0 \sim P_x$, independent of the training data X, define the out-of-sample prediction risk by

$$R_X(\hat{\boldsymbol{\beta}}; \boldsymbol{\beta}) := \mathbb{E}\left[(x_0^\top \hat{\boldsymbol{\beta}} - x_0^\top \boldsymbol{\beta})^2 \mid X\right] = \mathbb{E}\left[\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_\Sigma^2 \mid X\right].$$

Remark 1 (Bias–variance decomposition)

$$R_X(\hat{\boldsymbol{\beta}}; \boldsymbol{\beta}) = \underbrace{\|\mathbb{E}(\hat{\boldsymbol{\beta}} \mid X) - \boldsymbol{\beta}\|_\Sigma^2}_{B_X(\hat{\boldsymbol{\beta}};\boldsymbol{\beta})} + \underbrace{\mathrm{tr}\big(\mathrm{Cov}(\hat{\boldsymbol{\beta}} \mid X)\Sigma\big)}_{V_X(\hat{\boldsymbol{\beta}};\boldsymbol{\beta})}.$$

where

$$\|u\|_\Sigma^2 := u^\top \Sigma u.$$

Bias and Variance of the Min-Norm Estimator

Lemma 1 (Hastie et al. (2019), Lemma 1)

Under the model (1), (2) *the min-norm least squares estimator* $\hat{\beta}$ *has bias and variance*

$$B_X(\hat{\beta};\beta) = \beta^\top \Pi \Sigma \Pi \beta, \qquad V_X(\hat{\beta};\beta) = \frac{\sigma^2}{n}\,\mathrm{tr}(\hat{\Sigma}^+\Sigma),$$

where

$$\hat{\Sigma} = \frac{X^\top X}{n}$$

is the (uncentered) sample covariance of X*, and*

$$\Pi = I - \hat{\Sigma}^+\hat{\Sigma}$$

is the projection onto $\mathrm{null}(X)$*.*

High-Dimensional Regimes

Asymptotic aspect ratio

$$\gamma := \lim_{n,p\to\infty} \frac{p}{n}.$$

- **Underparameterized regime:** $\gamma < 1$
- **Interpolation threshold:** $\gamma = 1$
- **Overparameterized regime:** $\gamma > 1$

Why Does the Risk Blow Up Near Interpolation?

Theorem 2 (Hastie et al. (2019), Theorem 1)

Assume the model (1), (2) *and assume* $x \sim P_x$ *is of the form*

$$x_i = \Sigma^{1/2} z_i, \qquad i = 1, \ldots, n,$$

where z_i *is a random vector with i.i.d. entries that have zero mean, unit variance, and a finite* 4*th moment, and* Σ *is a positive definite matrix such that*

$$\lambda_{\min}(\Sigma) \geq c > 0$$

for all n, p *and a constant* c.
Assume also that the spectral distribution F_Σ *converges weakly to a measure* H. *Then as* $n, p \to \infty$, *such that*

$$\frac{p}{n} \to \gamma < 1,$$

the risk of the min-norm least-squares estimator $\hat{\beta}$ *satisfies, almost surely,*

$$R_X(\hat{\beta}; \beta) \to \sigma^2 \frac{\gamma}{1-\gamma}.$$

Why Does the Risk Blow Up Near Interpolation?

Definition 3

For a symmetric matrix $A \in \mathbb{R}^{p \times p}$, define its spectral distribution by

$$F_A(s) = \frac{1}{p} \sum_{i=1}^{p} \mathbf{1}\{\lambda_i(A) \leq s\},$$

where $\lambda_i(A)$, $i = 1, \ldots, p$, are the eigenvalues of A.

- The Marchenko–Pastur theorem implies that $F_{Z^\top Z/n}$ converges weakly to the law F_γ.
- In the isotropic case, the variance term satisfies

$$\frac{\sigma^2}{n} \operatorname{tr}(\hat{\Sigma}^{-1}\Sigma) = \frac{\sigma^2 p}{n} \int \frac{1}{s}\, dF_{Z^\top Z/n}(s) \longrightarrow \sigma^2 \gamma \int \frac{1}{s}\, dF_\gamma(s) = \sigma^2 \frac{\gamma}{1-\gamma}$$

- When $\gamma < 1$, then $p < n$ so that $X^\top X$ is invertible. it means that $\mathrm{null}(X) = \phi$. Thus, bias is zero.

Why Does the Risk Blow Up Near Interpolation?

Remark

In the underparameterized regime ($\gamma < 1$),

$$R_X(\hat{\boldsymbol{\beta}}; \boldsymbol{\beta}) = V_X(\hat{\boldsymbol{\beta}}; \boldsymbol{\beta}) \to \sigma^2 \frac{\gamma}{1-\gamma}.$$

- The risk is **purely variance**.
- It does not depend on the signal β or covariance Σ.
- As $\gamma \to 1^-$,

$$\sigma^2 \frac{\gamma}{1-\gamma} \to \infty,$$

  so the interpolation threshold is a **variance blow-up point**.

Overparameterized Regime: Limiting Bias

Now, we assume that $\Sigma = I$.

Lemma 4 (Hastie et al. (2019), Lemma 2)

Assume the model (1), (2), *where* $x \sim P_x$ *has i.i.d. entries with zero mean, unit variance, and a finite moment of order* $8 + \eta$ *for some* $\eta > 0$. *Assume that*

$$\|\beta\|_2^2 = r^2$$

for all n, p. *Then for the min-norm least squares estimator* $\hat{\beta}$, *as* $n, p \to \infty$ *such that*

$$\frac{p}{n} \to \gamma > 1,$$

its bias satisfies, almost surely,

$$B_X(\hat{\beta}; \beta) \to r^2 \left(1 - \frac{1}{\gamma}\right).$$

Bias as the Unrecoverable Signal Component

Recall the orthogonal decomposition

$$\boldsymbol{\beta} = \boldsymbol{\beta}_{\mathrm{row}} + \boldsymbol{\beta}_{\mathrm{null}}, \qquad \boldsymbol{\beta}_{\mathrm{row}} \in \mathrm{row}(X), \quad \boldsymbol{\beta}_{\mathrm{null}} \in \mathrm{null}(X).$$

Also, we have

$$\|\boldsymbol{\beta}\|_2^2 = \|\boldsymbol{\beta}_{\mathrm{row}}\|_2^2 + \|\boldsymbol{\beta}_{\mathrm{null}}\|_2^2.$$

In the isotropic case $\Sigma = I$,

$$B_X(\hat{\boldsymbol{\beta}}; \boldsymbol{\beta}) = \boldsymbol{\beta}^\top \Pi \boldsymbol{\beta},$$

where Π is the projection onto $\mathrm{null}(X)$. Hence

$$B_X(\hat{\boldsymbol{\beta}}; \boldsymbol{\beta}) = \|\Pi \boldsymbol{\beta}\|_2^2 = \|\boldsymbol{\beta}_{\mathrm{null}}\|_2^2.$$

Therefore, if

$$\|\boldsymbol{\beta}\|_2^2 = r^2,$$

then the bias is exactly the portion of the total signal energy lying in $\mathrm{null}(X)$.

Why the Limiting Bias Equals $r^2(1 - 1/\gamma)$

In the Gaussian special case $X_{ij} \overset{\text{i.i.d.}}{\sim} N(0,1)$, rotational invariance implies that X and XU have the same distribution for any orthogonal $U \in \mathbb{R}^{p \times p}$.

Since

$$B_X(\hat{\boldsymbol{\beta}}; \boldsymbol{\beta}) = \boldsymbol{\beta}^\top \Big(I - (X^\top X)^+ X^\top X \Big) \boldsymbol{\beta},$$

we may rotate $\boldsymbol{\beta}$ without changing its distributional behavior:

$$B_X(\hat{\boldsymbol{\beta}}; \boldsymbol{\beta}) \overset{d}{=} r^2 - (U\boldsymbol{\beta})^\top (X^\top X)^+ X^\top X (U\boldsymbol{\beta}).$$

Choose U so that $U\boldsymbol{\beta} = re_i$, then average over $i = 1, \ldots, p$:

$$B_X(\hat{\boldsymbol{\beta}}; \boldsymbol{\beta}) \overset{d}{=} r^2 \left(1 - \frac{1}{p} \operatorname{tr}\big((X^\top X)^+ X^\top X\big) \right).$$

Using

$$\operatorname{tr}\big((X^\top X)^+ X^\top X\big) = \operatorname{rank}(X) = n,$$

we obtain

$$B_X(\hat{\boldsymbol{\beta}}; \boldsymbol{\beta}) \overset{d}{=} r^2 \left(1 - \frac{n}{p} \right) \longrightarrow r^2 \left(1 - \frac{1}{\gamma} \right).$$

- The same limit continues to hold for isotropic non-Gaussian features; this is justified by a generalized Marchenko–Pastur theorem of Rubio and Mestre (2011).

Overparameterized Regime: Limiting Variance

Lemma 5 (Hastie et al. (2019), Lemma 3)

Assume the model (1), (2), *where* $x \sim P_x$ *has i.i.d. entries with zero mean, unit variance, and a finite* 4*th moment. For the min-norm least squares estimator* $\hat{\beta}$, *as* $n, p \to \infty$ *with*

$$\frac{p}{n} \to \gamma > 1,$$

its variance satisfies, almost surely,

$$V_X(\hat{\beta}; \beta) \to \frac{\sigma^2}{\gamma - 1}.$$

Overparameterized Regime: Limiting Risk

Theorem 6 (Hastie et al. (2019), Theorem 2)

Assume the model (1), (2), *where* $x \sim P_x$ *has i.i.d. entries with zero mean, unit variance, and a finite moment of order* $8 + \eta$, *for some* $\eta > 0$. *Also assume that*

$$\|\beta\|_2^2 = r^2$$

for all n, p. *Then for the min-norm least squares estimator* $\hat{\beta}$ *in* (4), *as* $n, p \to \infty$ *such that*

$$\frac{p}{n} \to \gamma > 1,$$

it holds almost surely that

$$R_X(\hat{\beta}; \beta) \to r^2 \left(1 - \frac{1}{\gamma}\right) + \frac{\sigma^2}{\gamma - 1}.$$

Thus, beyond interpolation, variance decreases faster while bias increases more slowly, producing the second descent.

Asymptotic Risk Curve

Remark 2

In the isotropic case, combining Theorems 2 and 5, the asymptotic risk of the min-norm least squares estimator can be written as

$$R(\gamma) = \begin{cases} \sigma^2 \dfrac{\gamma}{1-\gamma}, & \gamma < 1, \\ r^2 \left(1 - \dfrac{1}{\gamma}\right) + \dfrac{\sigma^2}{\gamma - 1}, & \gamma > 1. \end{cases}$$

Null Risk and SNR

Let $\tilde{\boldsymbol{\beta}} = \mathbf{0}$ be the null estimator. Then

$$R_X(\tilde{\boldsymbol{\beta}}; \boldsymbol{\beta}) = \mathbb{E}\left[(x_0^\top \boldsymbol{\beta})^2\right] = \boldsymbol{\beta}^\top \boldsymbol{\Sigma} \boldsymbol{\beta}.$$

In the isotropic case $\boldsymbol{\Sigma} = I$,

$$R_X(\tilde{\boldsymbol{\beta}}; \boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_2^2 = r^2.$$

Hence r^2 is the *null risk*, and

$$\mathrm{SNR} := \frac{r^2}{\sigma^2}$$

measures signal strength relative to noise.

Asymptotic Risk Curve

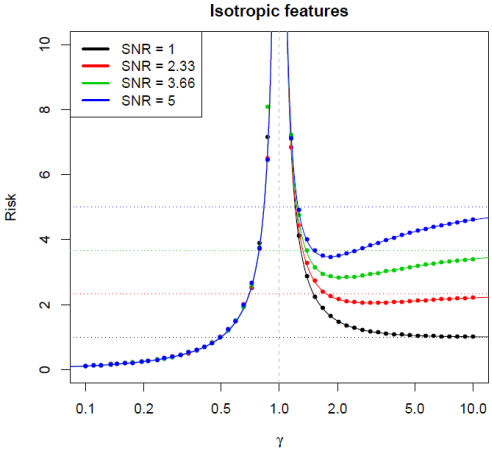

Figure 5: Asymptotic risk curves in the isotropic case with $r^2 \in \{1, 2.33, 3.66, 5\}$, $\sigma^2 = 1$, $n = 200$, $p = \lfloor \gamma n \rfloor$, and $X_{ij} \overset{\text{i.i.d.}}{\sim} N(0, 1)$. Dotted lines indicate the null risk.

Why Is This Relevant to Deep Learning?

- In the lazy / linearized regime, deep network training can resemble min-norm least squares with random features.
- The paper also studies a nonlinear random-feature model

$$x_i = \phi(Wz_i),$$

  where z_i is a Gaussian input vector, W is a random weight matrix, and ϕ is an activation function.
- The resulting risk curves are qualitatively similar to the linear isotropic theory.

Important caution

In real neural networks, the features are **learned**, whereas in this paper the features x_i are **observed** or fixed.

So this paper gives a clean mechanism, not the full story of deep learning.

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Beyond Parameter Count: Train Time Also Matters

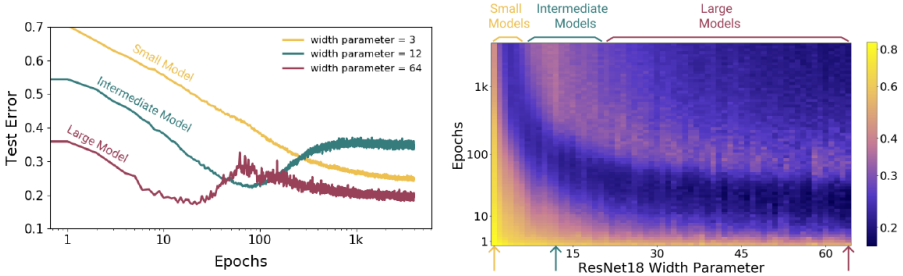

Figure 6: Epoch-wise double descent and the model size × epoch landscape

- Increasing train time can increase effective model complexity.
- A fixed architecture may move across regimes during training.
- The critical behavior again appears near the interpolation threshold.

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Point of Today's Results

- Classical bias–variance tradeoff does not fully explain modern generalization.
- In min-norm interpolation, double descent can be derived analytically.
- In deep learning, the same phenomenon extends beyond parameter count: model size and train dynamics both matter.

Take-home message

Interpolation is not the end of generalization. In modern learning, both overparameterization and training dynamics shape the test risk.

Interpolation vs. Extrapolation

- In the standard i.i.d. setup, training and test samples come from the same distribution $P(X, Y)$.
- Thus, test prediction is typically an **interpolation problem** within the support of the data distribution.
- By contrast, inputs far outside that support correspond to **extrapolation**.
- This is closer to distribution shift / OOD generalization than to ordinary test error analysis.
- Classical and modern generalization theory mostly study interpolation-type generalization, not true extrapolation.

Why Extrapolation Is Hard for Standard Deep Learning

- The training objective only constrains performance on the observed data distribution.
- Overparameterized neural networks are highly flexible outside the training support.
- Standard architectures do not automatically encode global rules such as conservation laws, symmetries, or causal structure.
- To improve extrapolation, current research introduces stronger structure: physics-informed neural networks, causal representation learning, and OOD generalization.

Large models alone are not enough. extrapolation needs structural bias.

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